Ishan Shah

Senior Software Engineer | Distributed Systems | Event-Driven Platforms

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ishandshah.github.io/portfolio/

Profile

I am a Senior Software Engineer working on distributed systems for billing, inventory, fulfillment, and security analytics. Across PayPal, Nordstrom, and Securonix, my work has centered on a practical question: how do we keep business state correct when services fail, events arrive twice, or a platform needs to recover?

Professional Experience

PayPal, Inc. | December 2024 - Present

Senior Software Engineer: May 26, 2025 - Present Previously MTS1 / Staff Software Engineer: December 2024 - May 25, 2025

The team delivered the platform to production in approximately six months. My contribution spanned service implementation, API design, performance engineering, operational readiness, and integration guidance.

Developed Java and Spring services for usage and subscription billing. Co-designed merchant APIs, asynchronous Kafka/MSK processing, multi-tenant controls, and recovery behavior. Contributed performance engineering, observability, runbooks, and merchant documentation.

Nordstrom, Inc. | Senior Software Engineer | March 2020 - December 2024

Design lead and platform owner for fulfillment optimization, partnering with Data Science and operational stakeholders. Led the rules-based store allocation engine and a CDC-based cost-accounting pipeline. Founding engineer on marketplace inventory infrastructure; designed partitioning changes and bounded Kafka recovery.

Securonix, Inc. | Software Engineer | December 2016 - March 2020

Developed ingestion, enrichment, and search infrastructure for security analytics. Extended Spotter query capabilities and optimized Solr routing and sharding. Contributed multi-cluster indexing, storage-tier query stitching, and SaaS historical-query paths.

Education

M.S., Computer Science | California State University, Los Angeles | 2014 - 2016

B.E., Computer Engineering | Gujarat Technological University | 2008 - 2013

Selected Engineering Outcomes

Fulfillment Optimization

Backtests and production experiments identified approximately 20-25% cost-reduction opportunities relative to the distance-only baseline, primarily through fewer express upgrades and late penalties. The v5 model ramped from 5% to 100% of targeted traffic over approximately three months.

Rules-Based Store Allocation

Processing dropped from approximately 9-12 minutes per purchase-order line to under 30 seconds per line. Peak backlogs of two to three days were replaced by processing close to purchase-order creation.

Marketplace Inventory Platform

The platform spanned more than 377 facilities and was designed for approximately 100,000 events per second. Those figures describe platform scope and design capacity, not a claim that every workload sustained that rate.

Kafka Partitioning and Recovery

Post-isolation measurements showed approximately 100% higher throughput for critical Transfer, RTV, Notification, and Supply Chain streams. The recovery worker replaced manual record identification and republishing with bounded, observable replay.

Oracle CDC and Cost Accounting

The pipeline enabled near-real-time item-level weighted-average cost, item-location stock-on-hand projections, and end-of-week accounting snapshots. It was designed for approximately 100,000 transactions per second; an inventory movement could generate more than 100 cost events.

SolrCloud Search Architecture

Solr performance improved approximately twofold through tuning, routing, and implicit sharding. Separately, the multi-cluster indexing design supported rates up to approximately 500,000 events per second. These are distinct performance results.

Technical Expertise

Java, Kotlin, Python, Spring Boot, Kafka/MSK, Kafka Streams, Debezium, PostgreSQL, ClickHouse, DynamoDB, Redis, Oracle, Solr, HDFS, AWS, Kubernetes, Docker, Terraform, OpenTelemetry, Gurobi, Optimizely.

Read the detailed engineering case studies

Publications, Speaking & Service

Technical Publications

Consumer Lag Is the Wrong Finish Line

DZone | 2026-10-13 | Ishan Shah

The Best AI Fix Isn't Adding More: It's Taking Away

HackerNoon | 2026-10-03 | Co-authored with @hacker86245963

Testing Kafka Consumer Recovery Contracts With Testcontainers and JUnit

Foojay | 2026-09-28 | Ishan Shah

AI Incident Response Needs A Control Plane, Not a Chatbot

HackerNoon | 2026-09-23 | Ishan Shah

Designing Replay-Safe CDC Pipelines With Kafka, Debezium, and Recovery Contracts

DZone | 2026-09-01 | Ishan Shah

AI Agents Should Bring Receipts, Not Just Answers

HackerNoon | 2026-08-17 | Ishan Shah

Building Idempotent Java Consumers for Replay-Safe Kafka Pipelines

Java Code Geeks | 2026-07-14 | Ishan Shah

Building Event-Driven Systems That Can Recover With Confidence

HackerNoon | 2026-06-27 | Ishan Shah

Completed Speaking

Conf42 Feature Flags (October 15, 2026); WeAreDevelopers North America (September 25); CloudX (September 3); KCD SF Bay Area (September 1); Zenity AI Agent Security Summit (May 27, co-presenter); AgentCon Raleigh (May 6).

Reviewing & Professional Service

Journal reviewer: IEEE Networking Letters, IEEE Transactions on Cloud Computing, IEEE Network, International Journal of Computers and Applications, and Journal of Cyber Security Technology. Technical book reviewer for Apress / Springer Nature; book-proposal reviewer for Manning.

Hackathon and award evaluation, technical mentoring, and startup advisory. Fellow Member, IICSPA (2026); Fellow, Hackathon Raptors (2025).

Professional service

Mentoring and advisory

Upcoming speaking is listed separately on the website